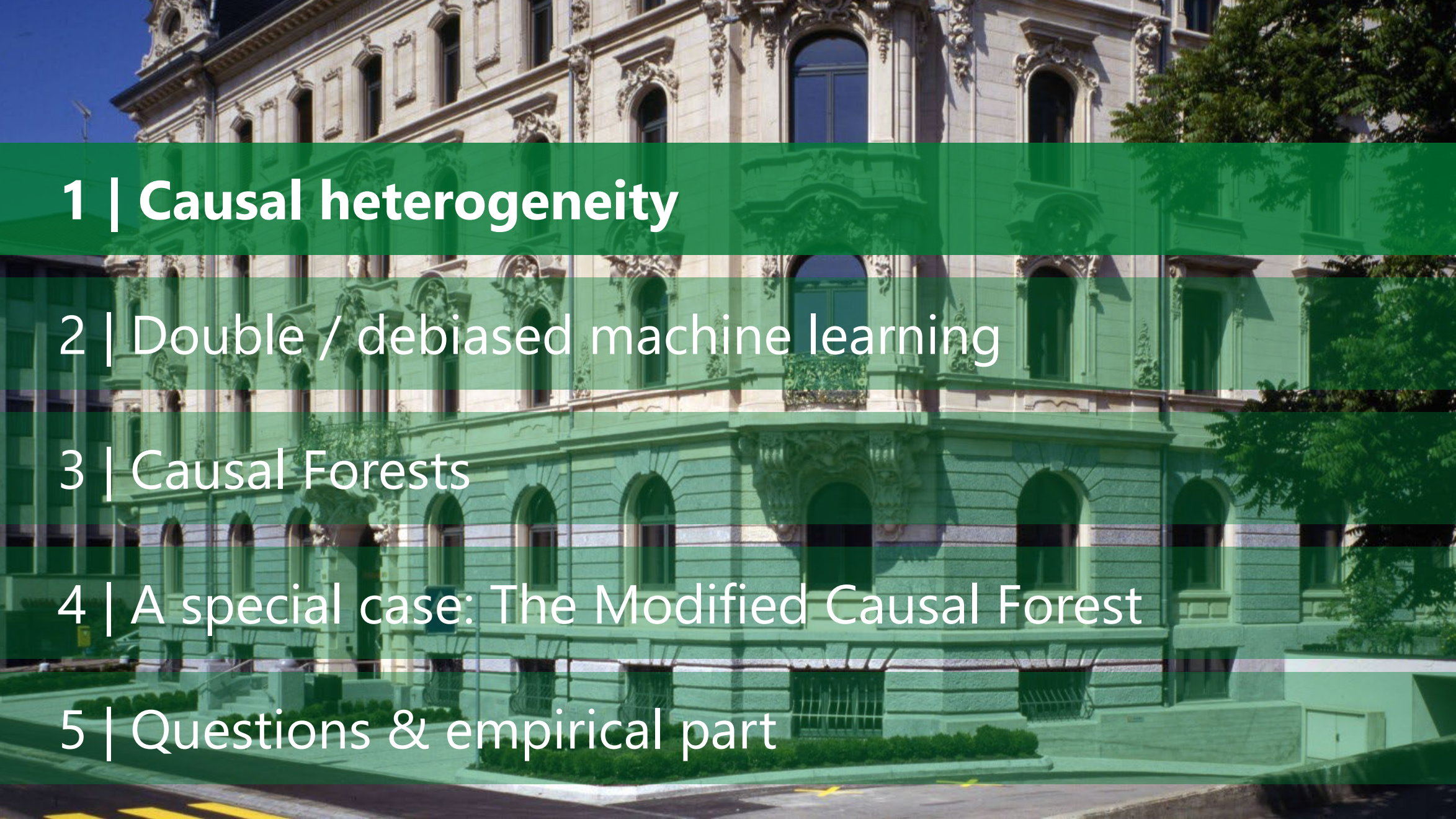
Causal Machine Learning & Artificial Intelligence for Economists

Michael Lechner
Swiss Institute for Empirical Economic Research (SEW-HSG)
University of St. Gallen | Switzerland

Lecture 4: Causal Machine Learning (Conditional Average Treatment Effects)

Required new external Python modules (from PyPI): mcf

Disclaimer: Almost all figures created by ChatGPT



1 | Causal heterogeneity

2 | Double / debiased machine learning

3 | Causal Forests

4 | A special case: The Modified Causal Forest

5 | Questions & empirical part



Why should economists care about heterogeneous effects?

Policies rarely affect everyone in the same way

Individuals differ in skills, constraints, preferences, & outside options

Average effects are useful, but policy questions quickly become: who gains most, who should receive it, and who may be harmed?

Efficiency
Allocate scarce resources where gains are largest

Equity
Detect who benefits, who does not, and who may be harmed

Design
Adapt duration, intensity, or versions of programmes



Average effects can hide important differences

A single average can mask winners and losers

- The same ATE can arise from very different subgroup effects

Interpretation
The policy may look mildly positive ***on average***, but this can hide large positive & negative effects

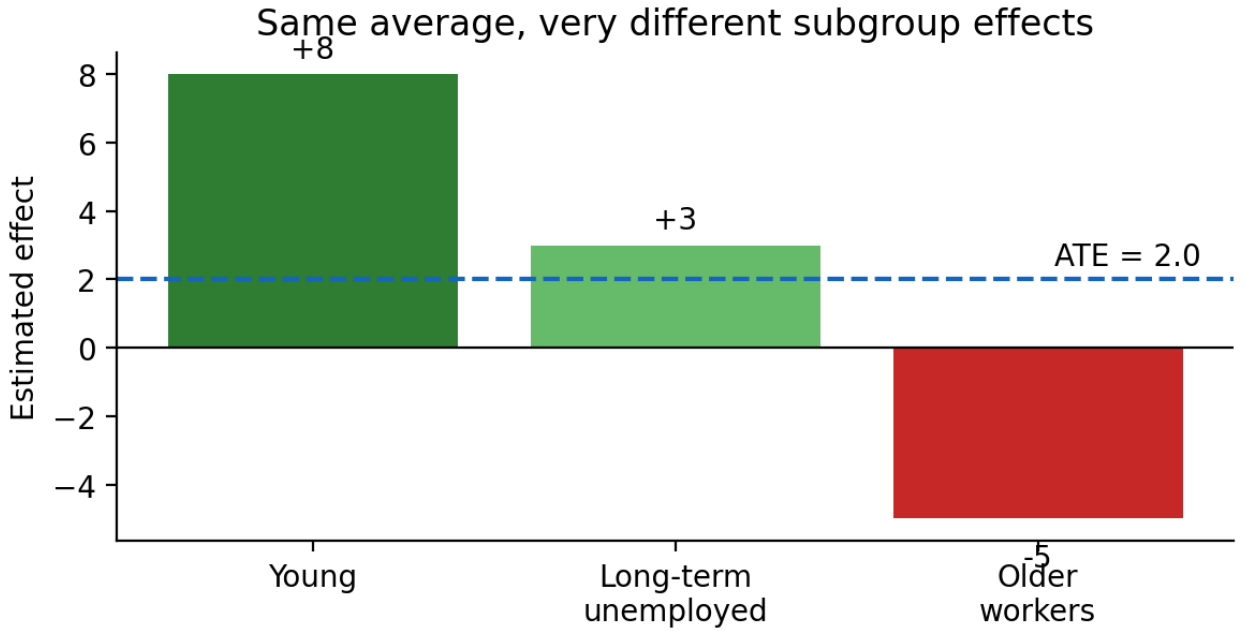




Illustration: one programme, different people

Consider a labour-market training programme

- Recent graduates may need only limited help
- Long-term unemployed may benefit strongly from intensive support
- Older workers may need a different programme version to gain much

A positive ATE does not tell us which group to prioritise

Group	Likely effect
Recent graduates	small to moderate
Long-term unemployed	large positive
Older workers	depends on version

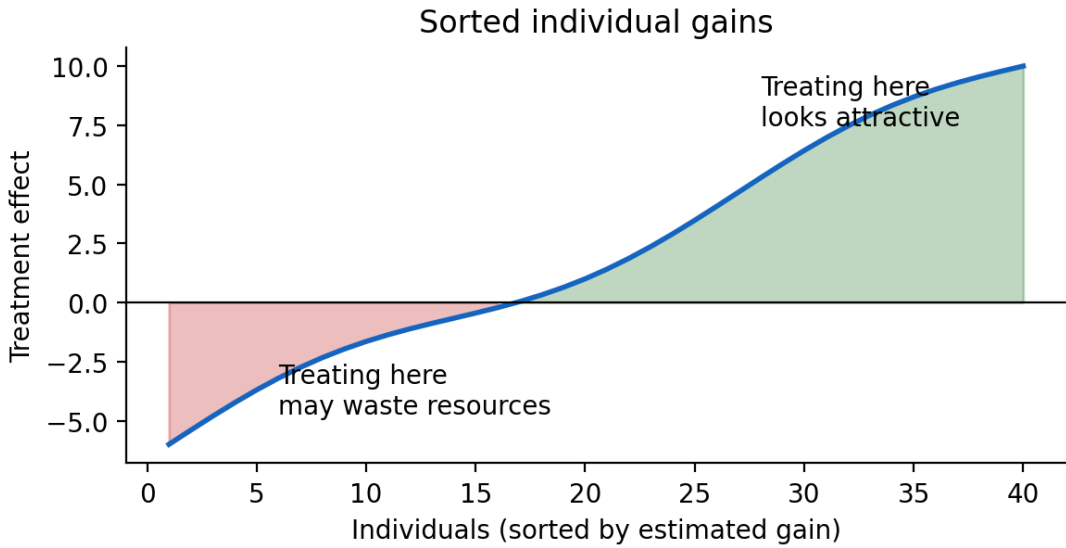


Why heterogeneity matters for targeting

If effects differ, a uniform assignment can waste resources

A planner may like to prioritise those with the largest expected gains, subject to constraints

Estimated effect heterogeneity is the foundation for data-driven policy targeting



Key message

Good policy needs heterogeneous treatment effects, not just population average effects



Good outcomes are not the same as large effects

A high expected outcome does not imply a large treatment gain

Risk profiling is therefore not the same as treatment-effect profiling

Person	Predicted Y(0)	Predicted Y(1)	Effect
A	0.80	0.82	+0.02
B	0.45	0.60	+0.15

B has the larger causal gain, but smaller outcome



From CATE(x) to GATE, BGATE, and IATE

CATE(x): treatment effect conditional on covariates $X = x$

Key message

These are all versions of the same idea: we average treatment effects at different levels of aggregation

CATE

General class of effects: conditional effect for a given covariate profile

GATE

Special CATE: Group average treatment effect for a meaningful subgroup

BGATE

Special CATE: Balanced comparison of group effects, holding covariate composition comparable

IATE

Special CATE: Individualised treatment effect: the most disaggregated effect estimate



Individualised **A**verage **T**reatment **E**ffect

Expected effect for units with specific values

of potentially many characteristics

D: Treatment (0 or 1)

$Y(1)$: Outcome when $D=1$

$Y(0)$: Outcome when $D=0$

X: Confounder & heterogeneity variables

Z: Specific heterogeneity variables (low dim.)

Observable: $X,\ Z,\ Y = DY(1) + (1-D)Y(0)$

Individualised (Conditional) Average Treatment Effects

$$IATE(x) = E(Y(1) - Y(0) \mid X = x)$$

Practical message

The more disaggregated the object, the harder estimation and inference become



Group Average Treatment Effects

D: Treatment (0 or 1)

Y^1: Outcome when $D=1$

Y^0: Outcome when $D=0$

X: Confounder & heterogeneity variables

Z: Specific heterogeneity variables (low dim.)

Observable: $X,\ Z,\ Y = DY^1 + (1-D)Y^0$

Average effect for a larger group of interest

Group (Conditional) Average Treatment Effects

$$GATE(z) = E(Y(1) - Y(0) \mid Z = z) = E_{X|Z=z} IATE(x)$$

Related effects

GATE of the Treated

$$GATET(z) = E(Y(1) - Y(0) \mid Z = z, D = 1) = E_{X|Z=z,D=1} IATE(x)$$

Practical message

The more disaggregated the object, the harder estimation and inference become.

Balanced GATE (BGATE; Bearth, Lechner, JBES, 2025)

$$BGATE(z, \tilde{X}) = E_{\tilde{X}} E(Y(1) - Y(0) \mid Z = z, \tilde{X} = \tilde{x}) = E_{\tilde{X}} E_{X|Z=z,\tilde{X}=\tilde{x}} IATE(x)$$



Why do we need specialised methods?

Effect heterogeneity may depend on many variables & complex interactions

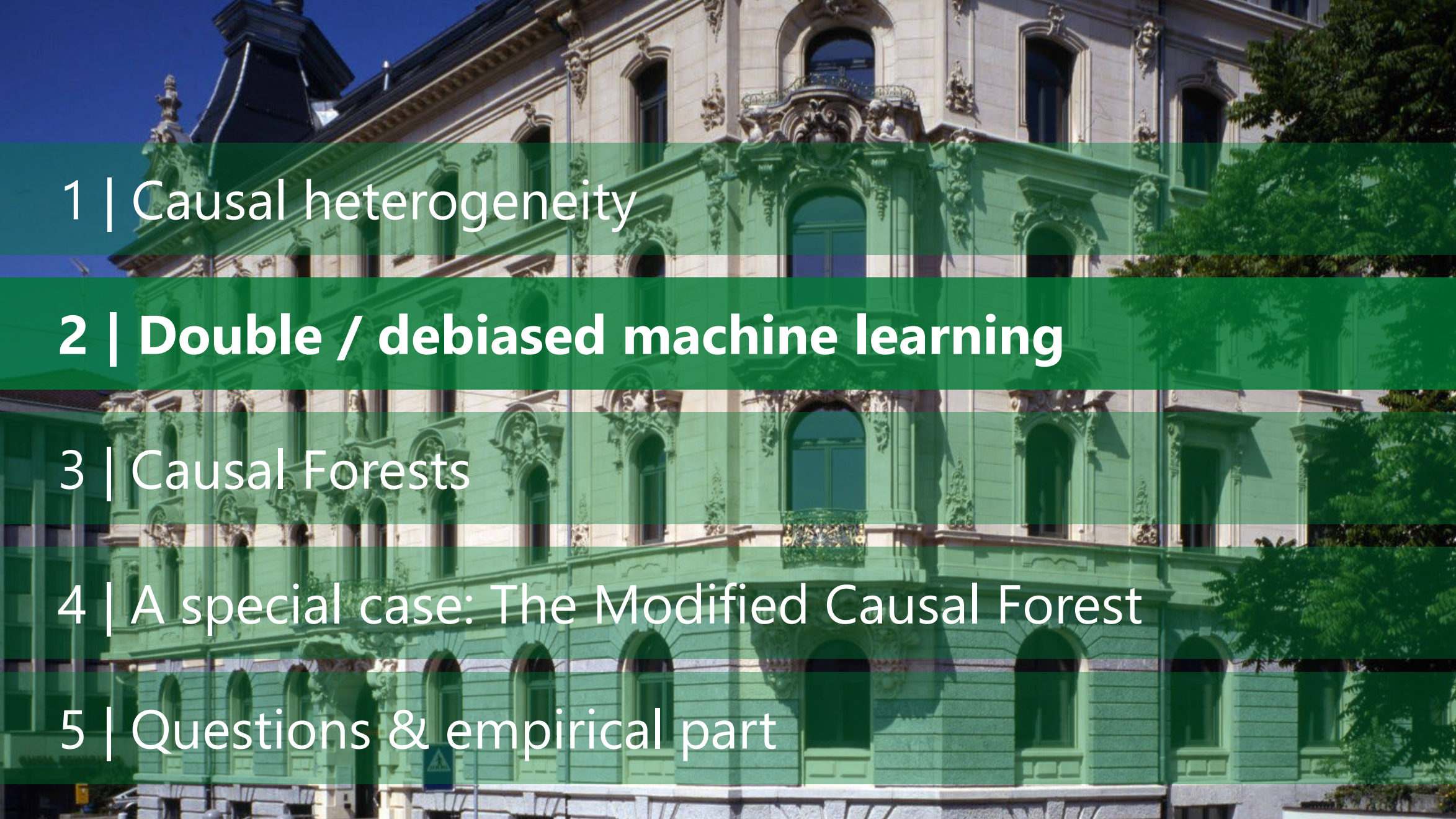
Ad hoc subgroup analysis quickly leads to multiple-testing & empty-cell problems

➔ We need flexible causal methods that can deal with these complications in a principled way (while respecting identification)

Main takeaway

Average effects are often too coarse for policy analysis & design

Next, we look at the tools available for getting heterogeneous effects:
DML for CATE estimation, Causal Forests in general, and the Modified Causal Forest in particular



1 | Causal heterogeneity

2 | Double / debiased machine learning

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The doubly robust score | 1

For estimating causal heterogeneity with DML, the **doubly robust score** is key (again)

$$drs(y_i, d_i, x_i) = \hat{\mu}_{-i}(1, x_i) - \hat{\mu}_{-i}(0, x_i) + d_i \frac{y_i - \hat{\mu}_{-i}(1, x_i)}{\hat{p}_{-i}(x_i)} - (1 - d_i) \frac{y_i - \hat{\mu}_{-i}(0, x_i)}{1 - \hat{p}_{-i}(x_i)}$$

- The nuisance functions $\hat{\mu}_{-1}(1, x_i), \hat{\mu}_{-1}(0, x_i), \hat{p}_{-1}(x_i)$ are estimated without using observation '*i*'
- Estimation of the *drs(y, d, x)* proceeds as discussed in the lecture on DML estimation of the ATE



Building estimators for GATEs & IATEs | 1

drs(y, d, x) depends on *y & d*, while effects depend only on features (*x, z*)

➔ take expectation of the *drs(y, d, x)* with respect to *Y & D* conditional on *X=x* (or Z=*z)*

Causal parameters may be of lower dimension than *drs*(..., *x*) w.r.t. *x*

➔ dimension reduction needed ➔ aggregate (=smooth)



Build estimators for GATEs & IATEs | 2

GATE(z)

- Z is discrete

  Average *drs*(y, d, x) in cells defined by the specific values of z ➔ *run OLS with drs(y, d, x) as dependent variable, & dummies with the values of z as independent variables*

- Z continuous:
  - (i) **OLS** with *drs(y, d, x)* as dependent variable, z as independent variables ➔ best linear predictor of *GATE(z)*
  - *(ii)* **Machine learner** with *drs(y, d, x)* as dependent variable, z as independent variables

IATE(x)

- (i) **OLS** with *drs(y, d, x)* as dependent variable, x as independent variables ➔ best linear predictor of *IATE(x)*
- *(ii)* **Machine learner** with *drs(y, d, x)* as dependent variable, x as independent variables



DML for binary treatment | Averages of DR score

ATE

$$\hat{ATE}_N = \frac{1}{N}\sum_{i=1}^{N} drs(y_i, d_i, x_i, \hat{\mu}_{-i}(\cdot), \hat{p}_{-i}(\cdot))$$

$$\sqrt{N}\frac{\hat{ATE}_N - ATE_0}{Var\left[drs(Y, D, X, g_0(\cdot), m_0(\cdot))\right]} \xrightarrow{d} N(0,1) \qquad \text{(as. efficient)}$$

GATE (z-discrete)

$$\hat{GATE}(z)_N = \frac{1}{N_z}\sum_{i=1}^{N} 1(z_i = z) drs(y_i, d_i, x_i, \hat{\mu}_{-i}(\cdot), \hat{p}_{-i}(\cdot)) \qquad N_z = \sum_{i=1}^{N} 1(z_i = z)$$

$$\sqrt{N_z}\frac{\hat{GATE}(z)_N - GATE_0(z)}{Var\left[drs(Y, D, X, g_0(\cdot), m_0(\cdot)) \mid Z = z\right]} \xrightarrow{d} N(0,1)$$

- Identical estimator: OLS with z-specific dummy variables
  - Use heteroscedasticity robust standard errors



DML for binary treatment | DR learner

GATE (z continuous) & IATE(x)

- If *f(x)* parametrically specified & estimation by OLS

$$\hat{CATE}(z)_N = \arg\min_{f(x)} \frac{1}{N} \sum_{i=1}^{N} \left[drs(y_i, d_i, x_i, \hat{\mu}_{-i}(\cdot), \hat{p}_{-i}(\cdot)) - f(z_i) \right]^2$$

$$\sqrt{N} \frac{\hat{CATE}(z)_N - f_0(z)}{E\left[\left[\hat{CATE}(z)_N - f_0(z)\right]^2\right]} \xrightarrow{d} N(0,1)$$

  - Drawback: Not robust to misspecification of *f(x)*
    - *If linear & OLS: Best linear predictor of CATE(x) (because of possible misspecification of f(z))*
- If Z continuous, or high-dimensional & *f(x)* is estimated nonparametrically, or by ML
  - Estimator is expected to be consistent under certain regularity conditions
  - Inference known only in special cases (e.g., continuous Z → nonparametric rate → Lechner, Zimmert, 2026)



Why consider alternative methods to *DML*?

Major difference to ATE estimator: DML is not necessarily semi-parametrically efficient

- This implies the possible existence of estimators that are better than DML

Lechner, Mareckova (2025) show severe biases & large variances of such DML estimators

- For *GATE*s when *z* has many values
- For *IATEs* more generally

Key message

DML gives us a way to estimate heterogeneous effects by first constructing a doubly robust score & then smoothing / aggregating that score appropriately.
However, there may be alternatives worth exploring …



1 | Causal heterogeneity

2 | Double / debiased machine learning

3 | Causal Forests

4 | A special case: The Modified Causal Forest

5 | Questions & empirical part



Recursive partitioning for heterogeneous causal effects

Susan Athey[a,1] and Guido Imbens[a]

[a]Stanford Graduate School of Business, Stanford University, Stanford, CA 94305

Causal Trees | *1*

Split data in homogeneous-in-*X* strata (leaf) →

→ Within a leaf, treated & control observations are more comparable in observed *X*

→ Under unconfoundedness, within-leaf outcome differences estimate leaf-specific causal effects

Recursive partitioning for heterogeneous causal effects

Susan Athey[a,1] and Guido Imbens[a]

[a]Stanford Graduate School of Business, Stanford University, Stanford, CA 94305

Causal Trees | 2

Splitting criteria: *max(Var(IATE(x)))*

Estimator of *IATEs(x)*: Difference of (weighted) within leaf treatment-control outcome means

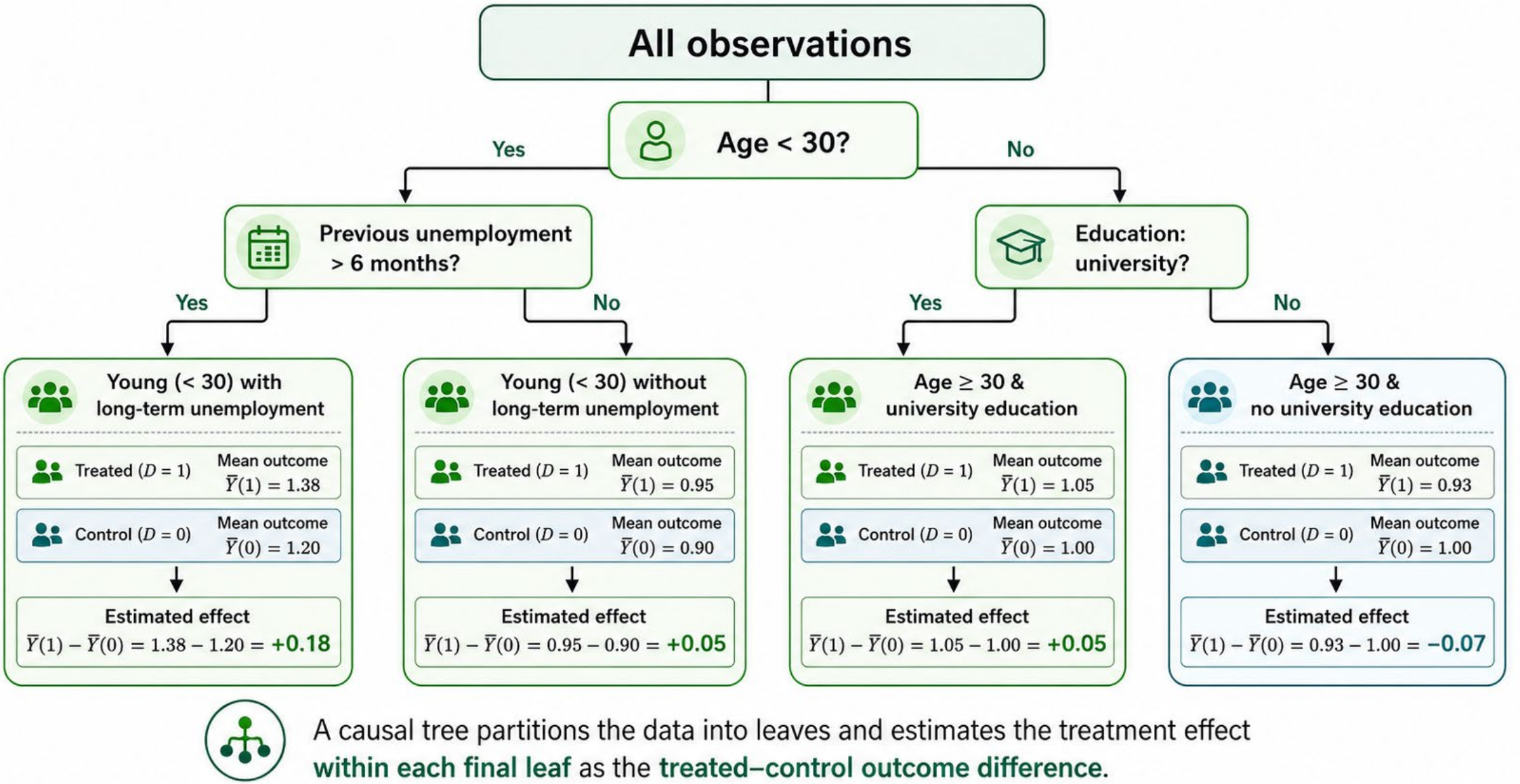
Honesty

- Different data to build forest & to populate the final leaves with y_i
- Avoidance of overfitting
- Debiasing & allowing for inference

Currently, rarely used for estimation

- Trees tend to be unstable & to have too much variance
- Outperformed by Causal Forests

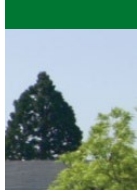
Causal Tree



A causal tree partitions the data into leaves and estimates the treatment effect **within each final leaf** as the **treated–control outcome difference.**



Journal of the American Statistical Association
ISSN: 0162-1459 (Print) 1537-274X (Online) Journal homepage: http://amstat.tandfonline.com/loi/uasa20



Basic principles of *Causal Forests* | 1

Estimation and Inference of Heterogeneous Treatment Effects using Random Forests

Stefan Wager & Susan Athey

Use deep causal trees

- Low bias, but high variance

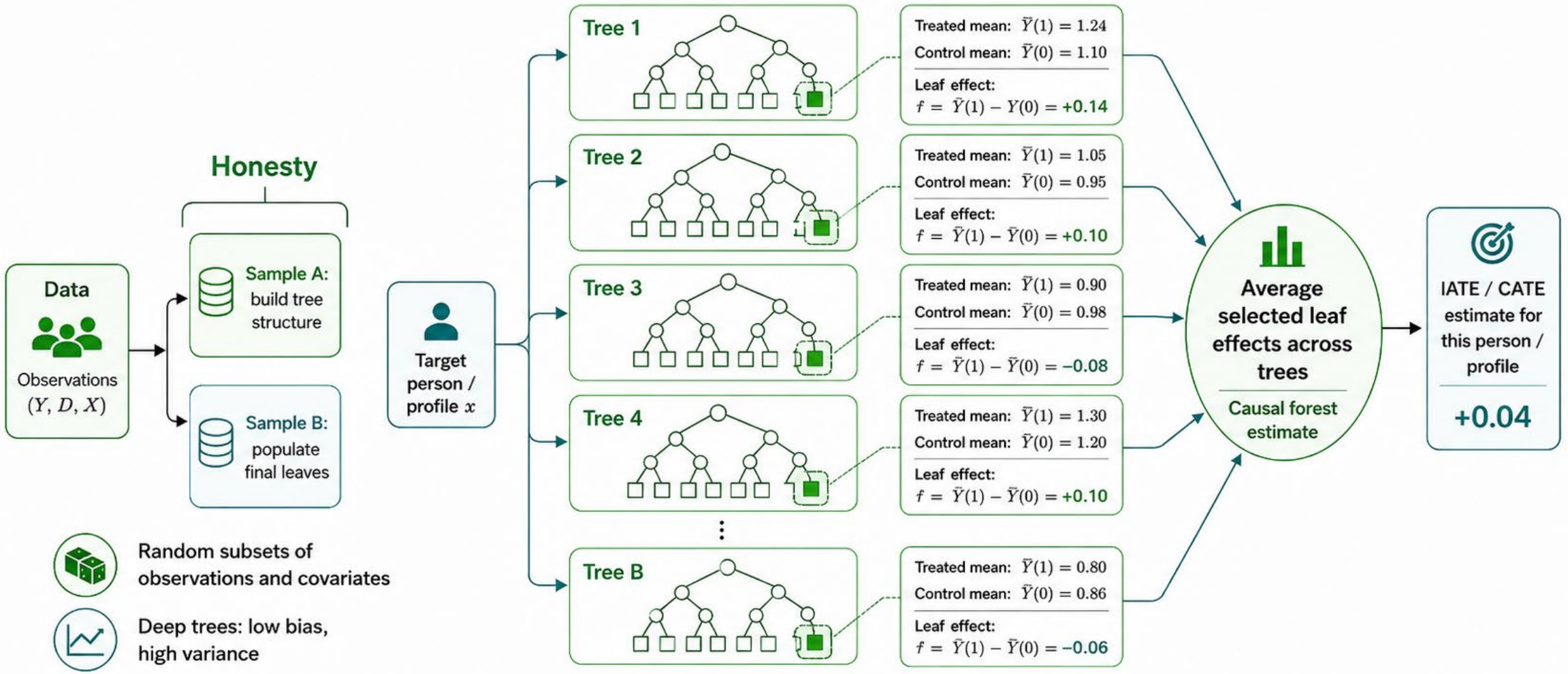
Use Random-Forest-like splitting algorithm with adapted splitting rule

- Different Causal Forests use different splitting rules that try to approximate the MSE of the IATE
  - Maximise treatment effect heterogeneity: CF (Wager, Athey, 2018), ***grf*** (Athey, Tibshirani, Wager, 2019)
  - Approximate estimation errors directly: *mcf* (Lechner, 2018)

Honest estimation to avoid bias & getting inference

- Different data to build forest & to populate the final leaves with y_i

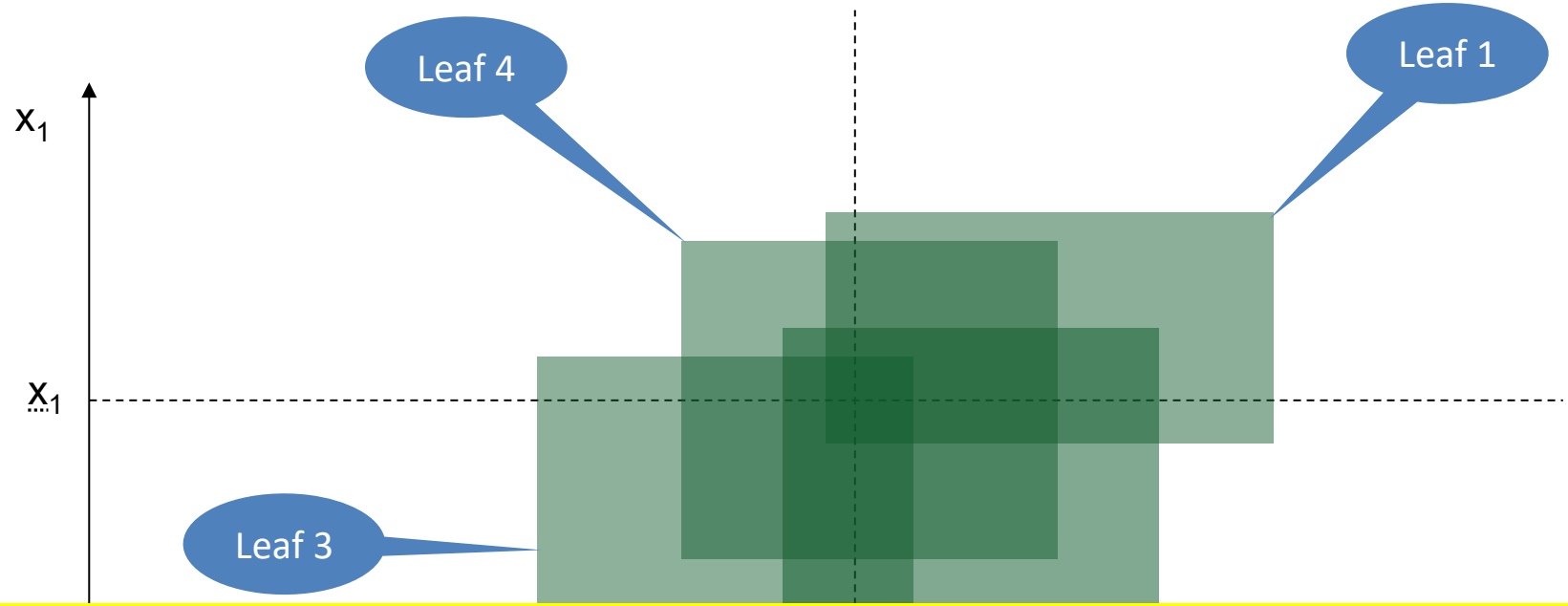
Causal Forest



A causal forest averages many **honest causal trees.** Each tree estimates the treatment effect within the terminal leaf as the **treated–control outcome difference.**

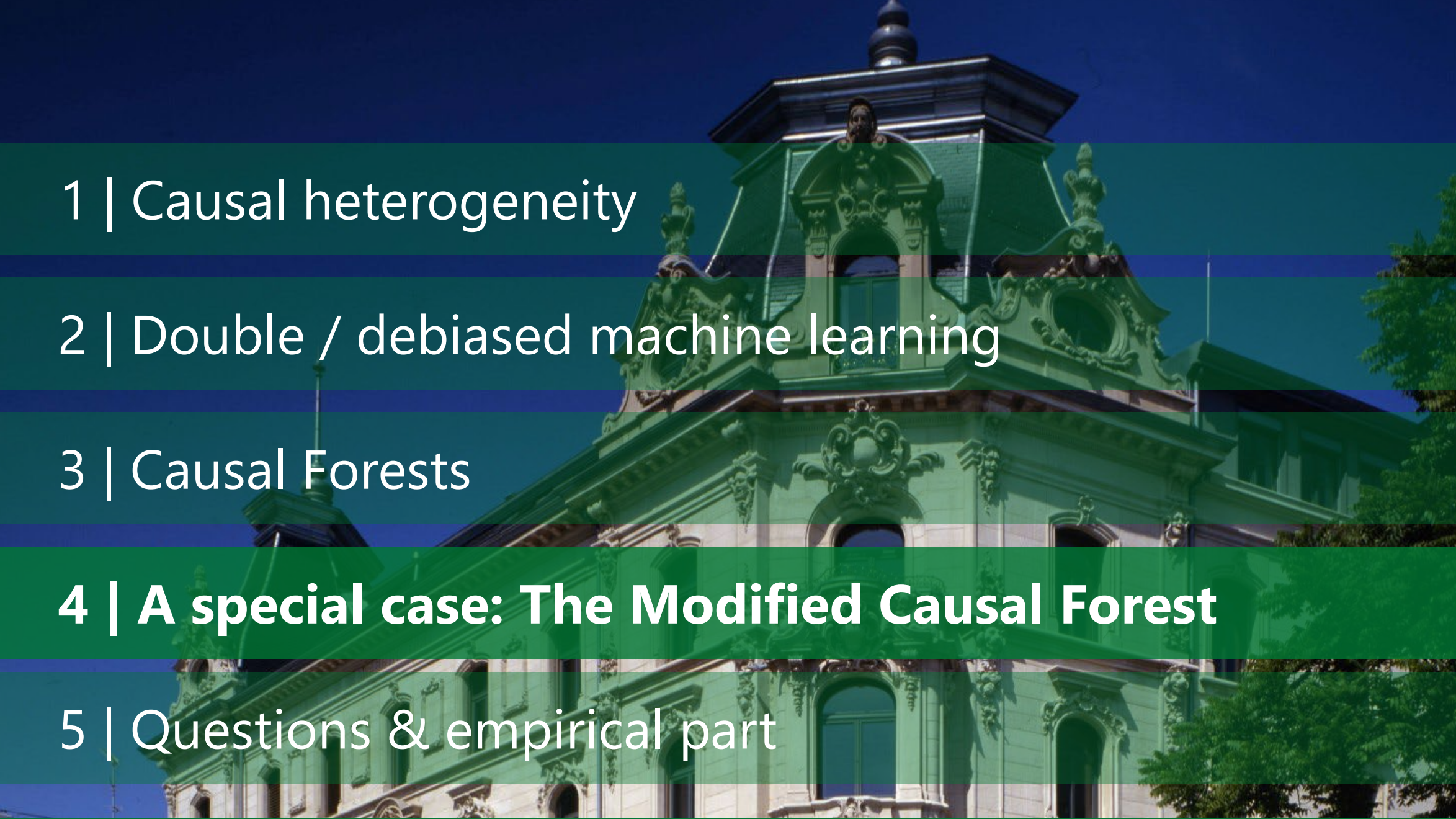


Bagged trees & Random Forests are weighting estimators of $f(x)$



Weighting:
(i) Prediction is average of means of y_i 's within each leaf
(ii) The closer x_i is to $\underline{x}$, the higher the probability that same y_i appears in many leaves
(iii) The weight of y_i for $f(\underline{x})=wy$ is proportional to # of occurrences of x_i in relevant leaves

Data adaptive size of leaves: Leaves have fixed number of observations ➔ the denser the data, the smaller the leaf (i.e., the smaller the implied bandwidth)



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The Applied Researchers' Wish List

Properties of estimator to be attractive in practice

Point estimators & inference for all effects

- All relevant aggregation levels
- Multiple treatments

Desirable statistical properties for reasonable *N*

- RMSE better than difference-of-regressions estimator
- Good basis for reliable inference: Smallish bias & normally distributed estimator
  - Maybe at the cost of some efficiency loss

Theoretical asymptotic guarantees for *point estimator* & *inference*

- Consistency, asymptotic normality for all parameters of interest
- Consistent estimator of standard errors / valid inference

Computationally not too demanding

- No separate estimations for different aggregation levels

IZA DP No. 12040
Modified Causal Forests for Estimating Heterogeneous Causal Effects

Modified **C**ausal **F**orest | Main ideas

Michael Lechner

DECEMBER 2018

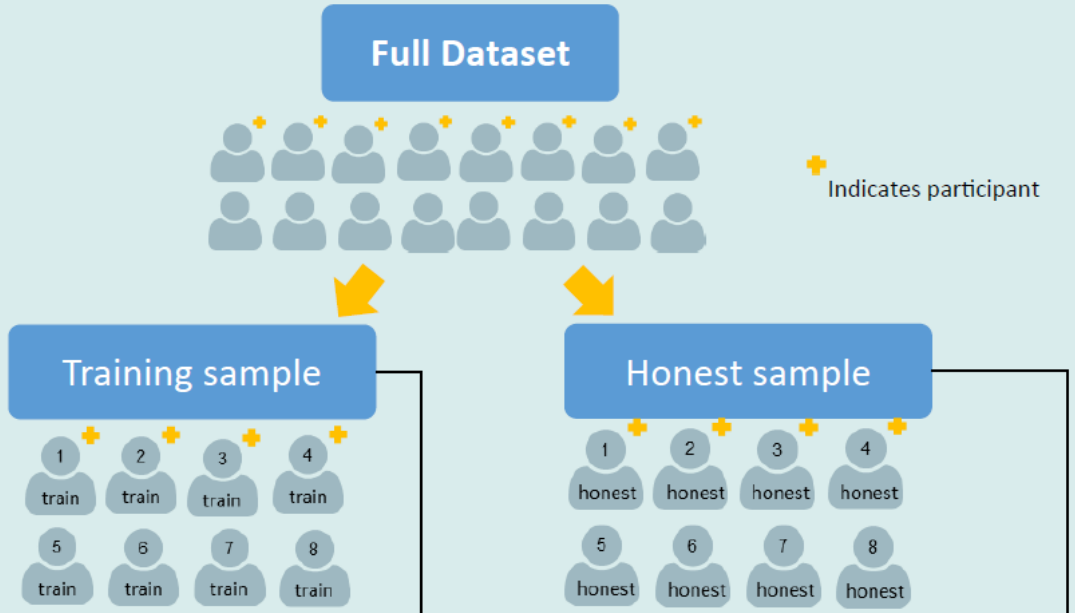
The *mcf* is a modification of the Causal Forest by Wager & Athey (2018)

Important modifications

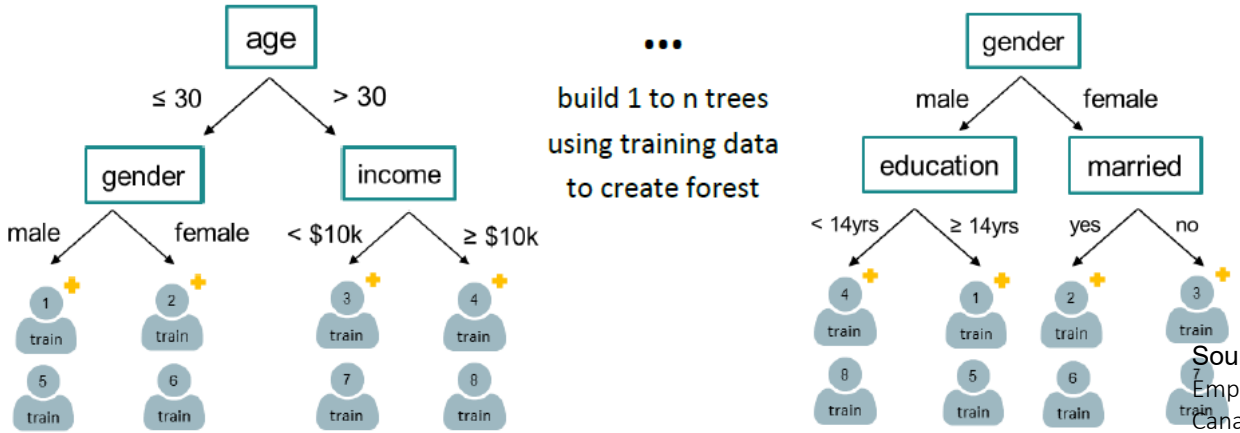
- Splitting rule in causal tree is based on alternative proxy for MSE
  - Nearest neighbour approximation of the MCE
- Penalty function to favour splits that lead to more extreme propensity scores in leaves
  - to put additional emphasis on reducing selection bias
- Representation of causal effect as weighted mean of y_i's is used for inference
  - Requires different implementation of sample splitting
- *All effects* are obtained by aggregating *IATE(x)*'s

Implementation of the Modified Causal Forest

Step 1: Split the estimation sample randomly into two equally-sized parts to create the training sample and the honest sample.



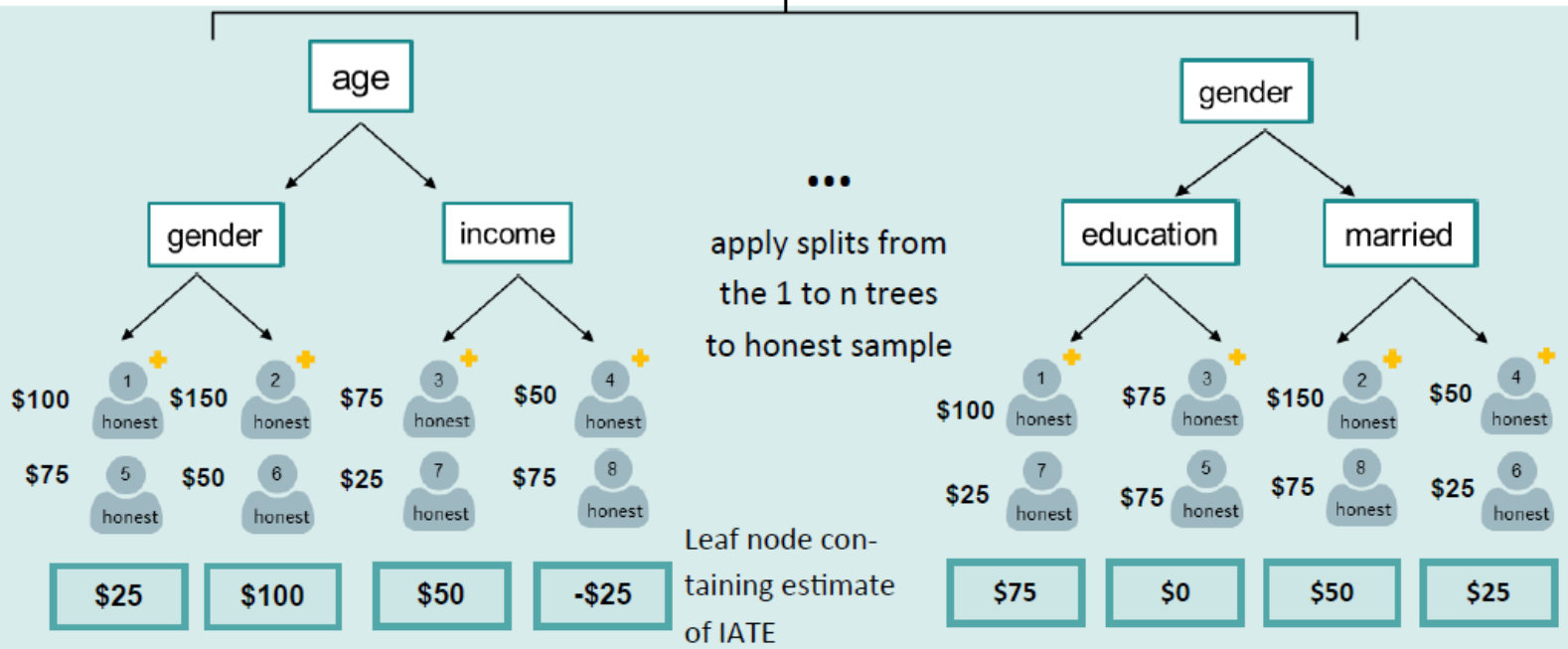
Step 2: Build the causal forest consisting of n trees using the training sample.



Source: Evaluation Directorate
Employment and Social Development Canada / Government of Canada

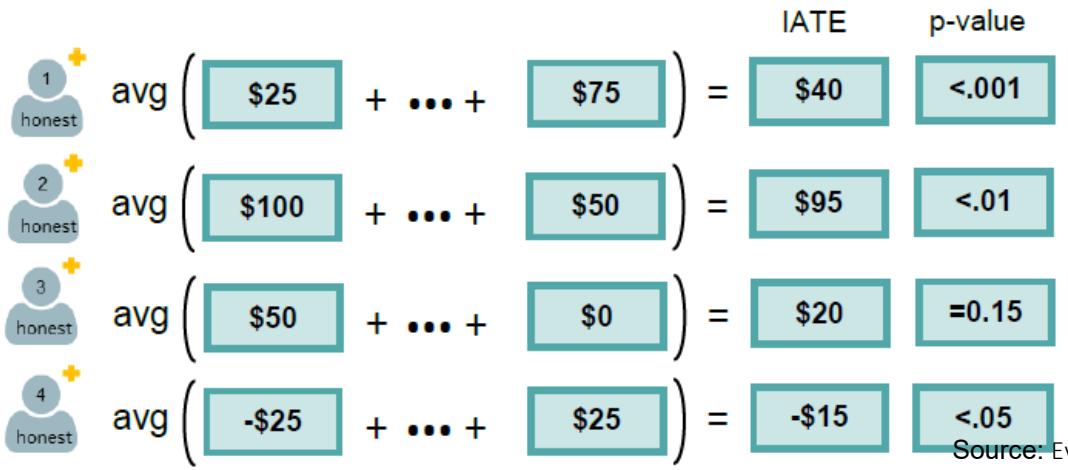
Step 3: Apply the splits obtained in the training sample to the honest sample.

Take the mean of the outcomes in the respective leaf as the estimate of the net impact. Repeat over number of trees that comes with this forest to obtain individualized net impacts.



Step 4: The individualized "average" treatment effect for each individual averages the individualized treatment effect from each of the n trees.

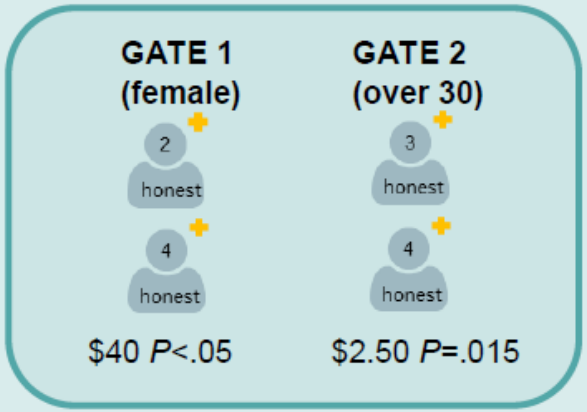
Inference on the IATEs is performed by obtaining weights from the forest, by counting how many times an observation in the honest sample is used to predict IATE, and estimating the variance using these weights.

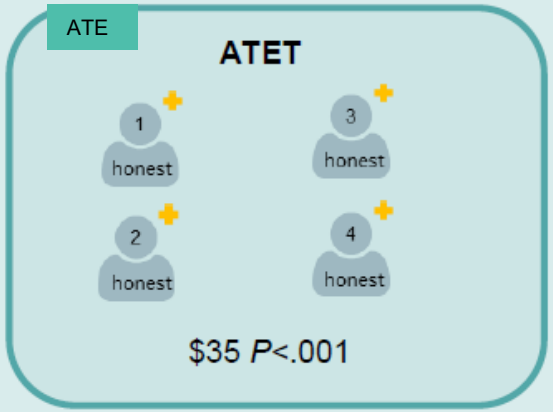


Source: Evaluation Directorate Employment and Social Development Canada / Government of Canada



Step 5: Aggregate the IATEs to GATEs or ATEs by taking the average over observations in the honest sample over a specific characteristics (e.g. females) or over all observations in the treatment group. Obtain the new weights for the GATE and ATEs to perform valid statistical inference.





Source: Evaluation Directorate Employment and Social Development Canada / Government of Canada



Modified Causal Forest | 3 | Asymptotic properties

ATE & GATE & IATE

- Consistent, asymptotically normal
  - under regularity conditions

ATE:
$$\frac{\hat{\theta}^{mcf} - \theta^0}{\sqrt{Var\left(\hat{\theta}^{mcf}\right)}} \xrightarrow{d} N(0,1)$$

GATE(z):
$$\frac{\hat{\theta}^{mcf}(z) - \theta^0(z)}{\sqrt{Var\left(\hat{\theta}^{mcf}(z)\right)}} \xrightarrow{d} N(0,1)$$

IATE(x):
$$\frac{\hat{\theta}^{mcf}(x) - \theta^0(x)}{\sqrt{Var\left(\hat{\theta}^{mcf}(x)\right)}} \xrightarrow{d} N(0,1)$$

arXiv > econ > arXiv:2405.10198

Economics > Econometrics

[Submitted on 16 May 2024]

Comprehensive Causal Machine Learning

Michael Lechner, Jana Mareckova

University of St.Gallen

Swiss Institute for
Empirical Economic Research



Causal machine learning informs government policies

MCF became default method to evaluate (& improve)

Canadian ALMP

- Government of Canada (ESDC)

UNICEF is capacity building for using *MCF* in policy

- Kazakhstan, Kosovo, …



Proceedings of Statistics Canada Symposium 2022:
Data Disaggregation: building a more representative data portrait of society

Heterogeneous causal effects of labour market programs: A machine learning approach

by Andy Handouyahia, Tristan Rikhi, Georges Awad and Essolaba Aouli

Release date: June 28, 2024

Statistics Canada Statistique Canada

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unicef | for every child

Office of Innovation

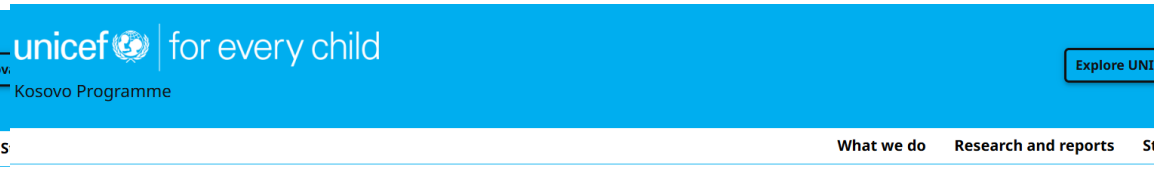
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Innovation Across Borders: Kazakhstan

José Sierra Castillo, UNICEF Europe and Central Asia, tells us about his experience testing a machine learning model for policy design in Kazakhstan



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Press release

Kosovo Institutions explore Machine Learning applications for evidence-based policymaking

05 November 2025







1 | Causal heterogeneity

2 | Double / debiased machine learning

3 | Causal Forests

4 | A special case: The Modified Causal Forest

5 | Questions & empirical part







Rules for group work (including graded group exercise)

Groups may have up to 4 members

- *All group members must agree to all group members*

Group memberships may change until lecture 3

You are free to use any AI you like

- Graded exam: Contact with humans is prohibited & will be monitored (& will be useless )

Lecturer is available for discussions & help during group work

- limited in graded group work

You should always ***install & test*** the required packages ***before the lecture***

- ***Required (additional) packages are always shown on 1st slide of each lecture***



Empirical group work | 1

Today, we use *ALMP_data.csv* for the effect estimation of active labour market programmes. You find additional information about the programmes in the file *ALMP_inst_info.pdf*.

Although there are four different programme types, we aggregate them into one group, i.e., the goal is to estimate the effect of participating in the active labour market programme in global.

As outcomes consider earnings & being employed in the 4th quarter of 19X9



Empirical group work | 2

All estimations are based on the *mcf* package

Estimate the ATE of participating in the ALMP

- Do they differ in a relevant way from the ones estimated by DoubleML in the last lecture?

Estimate the GATEs(sex)

- Do they differ in a relevant way from the ones estimated by DoubleML in the last lecture?

Estimate the GATEs for vocational education, age & nationality?

- Do you detect relevant heterogeneity?



Empirical group work | 3

Do the estimated IATEs indicate relevant effect heterogeneity?

- Use the sorted IATEs & the distribution of the IATEs to address this question provided by the analysis method of mcf

Are there indications of other features related to relevant effect heterogeneity?

- Use the k-means algorithm provided by the analysis method of mcf



Next lecture

Optimal decisions (optimal policy)



Background slides

(for interested students only, not for exam)



The estimation problem



Estimation of $IATE(x) = \mu(1;x) - \mu(0;x)$

Causal problem is transformed into prediction problem(s)

- Machine learning (ML) methods are good with prediction problem(s)

Goal is to estimate a *difference of* ***2*** conditional expectations in ***2*** (nonrandom) *populations* defined by observed treatment status

- Nonstandard ML problem
- 'Ground truth' not observable ('true' dependent variable unobservable)

Selection into these (sub-)populations by 'selection process'

- *Propensity score,* $p_m(x)$*,* affects properties of estimators by changing the …
  - observed joint distribution of *X* & *Y* in the subpopulations (confounding)
  - sample sizes of subpopulations (bias & precision)



General remarks on predicting IATE(x)

The ML *goodness-of-fit perspective* on this problem

$$\hat{IATE}_N(x) = \hat{\mu}(1;x) - \hat{\mu}(0;x); \qquad IATE(x) = \mu(1;x) - \mu(0;x)$$

$$\begin{aligned} MSE(\hat{IATE}_N(x)) &= E(\hat{IATE}_N(x) - IATE(x))^2 \\ &= E(\hat{\mu}(1;x) - \mu(1;x))^2 + E(\hat{\mu}(0;x) - \mu(0;x))^2 \\ &\quad - 2E(\hat{\mu}(1;x) - \mu(1;x))(\hat{\mu}(0;x) - \mu(0;x)) \\ &= MSE(\hat{\mu}(1;x)) + MSE(\hat{\mu}(0;x)) - 2MCE(\hat{\mu}(1;x), \hat{\mu}(0;x)). \end{aligned}$$

The last term (**M**ean **C**orrelated **E**rror) cannot be easily estimated

- If x is continuous, y_i's in treatments *1* **&** 0 with same value of x will not be observed
- Different estimation methods approximate MCE() differently



Remarks on this prediction problem | 3

$$MSE\left[\hat{I}ATE_N(x)\right] = MSE\left[\hat{\mu}(1;x)\right] + MSE\left[\hat{\mu}(0;x)\right] - 2MCE\left[\hat{\mu}(1;x), \hat{\mu}(0;x)\right]$$

The 2 estimators of the 2 regression curves are tied together

- Positively correlated estimation errors of the regression curves matter less

MCE leads to biased estimation of the 2 regression curves

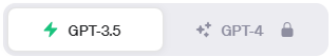
- If estimation errors are correlated across treatments, this will be a better estimator for the difference despite being a worse estimator for the 2 regression curves

ML principles may be modified

- to account for this 'tying together' of the regression curves
  - Different estimators use different approximations of the MCE()
- to remove bias of the estimator (needed for inference)



More details on the MCF



ChatGPT



Background

Objective function | MCE | 2

Optimise sample splits in training sample w.r.t.

$$MSE\left[\hat{\mu}(1;x)\right]+MSE\left[\hat{\mu}(0;x)\right]-2MCE\left[\hat{\mu}(1;x),\hat{\mu}(0;x)\right]$$

$$\hat{MSE}\left[\hat{\mu}_d(x)\right]=\frac{1}{N_{S_x}^d}\sum_{i=1}^{N}\underline{1}(x_i\in S_x)\underline{1}(d_i\in d)\left[\overline{y}_{S_x,d}-y_i\right]^2,\quad x\in S_x,\qquad \tilde{y}_{(i,m)}=\begin{cases} y_i & d_i=m \\ y_{(i,m)} & if \quad d_i\neq m\end{cases}$$

$$\hat{MCE}(x)=\frac{1}{N}\sum_{i=1}^{N}\underline{1}(x_i\in S_x)\left[\overline{y}_{S_x,1}-\tilde{y}_{(i,1)}\right]\left[\overline{y}_{S_x,0}-\tilde{y}_{(i,0)}\right]$$

Unknown counterfactuals substituted by nearest neighbours using main diagonal of Mahalanobis distance

- Matching on the propensity score(s) is a bad idea: MCE is not (directly) related to selection bias
- Matching on predictive scores works: Computationally more expensive than Mahalanobis matching as predictive scores need to be estimated by ML



Background

Objective function | Penalty term

Main idea

- Add penalty term that penalises splits with similar treatment probabilities
  - Reduces selection bias effectively

Example of a penalty function

$$penalty(x', x'') = \lambda \left\{ 1 - \frac{1}{M} \sum_{d=0}^{M-1} (P(D = d \mid X \in leaf(x') - P(D = d \mid X \in leaf(x''))^2 \right\}, \quad \lambda \geq 0$$

- = *0* if split leads to a perfect prediction of the probabilities in the daughter leaves
- = λ if treatment probabilities in daughter leaves are the same
- Cheap to compute as it only counts the treatments in any potential split



Background

Weights for aggregation & inference | 1

Estimators coming from RF/CF can be expressed as weighted outcome means

Estimators for intermediate levels can be obtained by aggregating weights

- Computationally convenient

Inference

- Weight-based inference
  - Same principle for all aggregation levels
  - Sample splitting required



Background

Inference | General weight-based estimators | 1

Structure of estimator: $\hat{\theta} = \frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i; \qquad Var(\hat{\theta}) = Var\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i\right).$

Law of total variance (*Var(Y)=E[Var(Y|X)]+Var[E(Y|X)]*):

$$Var\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i\right) = E_{\hat{W}} Var\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i \mid \hat{w}_1,...,\hat{w}_N\right) + Var_{\hat{W}} E\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i \mid \hat{w}_1,...,\hat{w}_N\right)$$

i.i.d. sampling:

$$Var\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i \mid \hat{w}_1,...,\hat{w}_N\right) = \frac{1}{N^2}\sum_{i=1}^{N} \hat{w}_i^2 Var(y_i \mid \hat{w}_1,...,\hat{w}_N)$$

$$E\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i \mid \hat{w}_1,...,\hat{w}_N\right) = \frac{1}{N}\sum_{i=1}^{N} \hat{w}_i E(y_i \mid \hat{w}_1,...,\hat{w}_N)$$

RF: $$Var\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i y_i\right) = E_{\hat{W}}\left(\frac{1}{N^2}\sum_{i=1}^{N} \hat{w}_i^2 Var(y_i \mid \hat{w}_1,...,\hat{w}_N)\right) + Var_{\hat{W}}\left(\frac{1}{N}\sum_{i=1}^{N} \hat{w}_i E(y_i \mid \hat{w}_1,...,\hat{w}_N)\right).$$

$\hat{w}_i = \hat{w}(x_i, \vec{X}^T, \vec{Y}^T)$ $(\vec{X}^T, \vec{Y}^T)$ are the training data to build RF



Background

Inference | General weight-based estimators | 2

If i does not belong to training data, and observations are i.i.d., y_i is independent of $\hat{w}_j$

$$E(y_i \mid \hat{w}_1, \ldots, \hat{w}_N) = E(y_i \mid \hat{w}_i) = \mu_{Y|W}(\hat{w}_i)$$

$$Var(y_i \mid \hat{w}_1, \ldots, \hat{w}_N) = Var(y_i \mid \hat{w}_i) = \sigma^2_{Y|W}(\hat{w}_i)$$

This leads to a simpler formula for the variance:

$$Var\left(\frac{1}{N}\sum_{i=1}^{N}\hat{w}_i y_i\right) = E_{\hat{W}}\left(\frac{1}{N^2}\sum_{i=1}^{N}\hat{w}_i^2 \sigma^2_{Y|\hat{W}}(\hat{w}_i)\right) + Var_{\hat{W}}\left(\frac{1}{N}\sum_{i=1}^{N}\hat{w}_i \mu_{Y|\hat{W}}(\hat{w}_i)\right).$$



Background

Inference | General weight-based estimators | 3

$$Var(\hat{\theta}) = E\Big(\Big[Var(\hat{\theta}\,|\,\hat{W}\Big]\Big) + Var\Big[E(\hat{\theta}\,|\,\hat{W})\Big] \qquad \text{Law of total variance}$$

… suggests to use the following estimator for the variance:

$$\widehat{Var(\hat{\theta})} = \frac{1}{N^2}\sum_{i=1}^{N}\hat{w}_i^2\hat{\sigma}^2_{Y|\hat{W}}(\hat{w}_i) + \frac{1}{N(N-1)}\sum_{i=1}^{N}\left[\hat{w}_i\hat{\mu}_{Y|\hat{W}}(\hat{w}_i) - \frac{1}{N}\sum_{i=1}^{N}\hat{w}_i\hat{\mu}_{Y|\hat{W}}(\hat{w}_i)\right]^2$$

These are 2 1-dimensional problems. Therefore, standard non-parametric estimators are fast & accurate for $\hat{\mu}_{Y|\hat{W}}(\hat{w}_i)$ & $\hat{\sigma}^2_{Y|\hat{W}}(\hat{w}_i)$

- Some estimation noise will be washed out by averaging these non-parametric estimators



Background

Inference | General weight-based estimators | 4

Implementation

- *k*-nearest neighbour estimators (*k* increases slowly with *N*)
  - Computationally convenient (fast)
  - Alternatively, Nadaraya-Watson estimation
- Bodory, Camponovo, Huber, & Lechner (2020) investigate *k*-nearest neighbour estimators to obtain these weights
  - they found good results in a binary treatment setting for the ATET
- Finite sample properties of different versions not yet investigated

Clustering can also 'easily' be incorporated

Shortcoming: Lack of theory for choosing smoothing parameters



Background

Inference | General weight-based estimators | 5

Final remarks on inference

- Would be nice to have some asymptotic theory for the estimators of conditional means & variances to get consistency conditions (problem a bit more non-standard as weights come from a machine learner)
- The more spread-out the weights are (as for ATE & GATEs), the better the inference, as it may not require consistent estimation of $\hat{\mu}_{Y|\hat{W}}(\hat{w}_i)$, $\hat{\sigma}^2_{Y|\hat{W}}(\hat{w}_i)$
- With the less spread-out weights of IATE, the estimator will much more benefit from good estimation of $\hat{\mu}_{Y|\hat{W}}(\hat{w}_i)$, $\hat{\sigma}^2_{Y|\hat{W}}(\hat{w}_i)$
- Open issue: Are the choices for *k* also reasonable for IATE (that removes less noise by averaging), or do we need larger or smaller *k*?



Background

Estimator of (G)ATE as function of estimator of IATE

$$\widehat{GATE}(m,l;z,\Delta) = \frac{1}{N^{z,\Delta}} \sum_{i=1}^{N} \underline{1}(z_i = z, d_i \in \Delta) \widehat{IATE}(m,l;x_i)$$

$$= \frac{1}{N^{z,\Delta}} \sum_{i=1}^{N} \underline{1}(z_i = z, d_i \in \Delta) \frac{1}{N} \sum_{j=1}^{N} \hat{w}_j^{IATE(m,l;x_i)} y_j$$

$$= \frac{1}{N} \sum_{i=1}^{N} \left(\frac{1}{N^{z,\Delta}} \sum_{j=1}^{N} \underline{1}(z_j = z, d_j \in \Delta) \hat{w}_i^{IATE(m,l;x_j)} \right) y_i = \frac{1}{N} \sum_{i=1}^{N} \hat{w}_i^{GATE(m,l;z,\Delta)} y_i;$$

$$\hat{w}_i^{GATE(m,l;z,\Delta)} = \frac{1}{N^{z,\Delta}} \sum_{j=1}^{N} \underline{1}(z_j = z, d_j \in \Delta) \hat{w}_j^{IATE(m,l;x_i)}; \qquad N^{z,\Delta} = \sum_{i=1}^{N} \underline{1}(z_i = z, d_i \in \Delta).$$

$$\widehat{ATE}(m,l;\Delta) = \frac{1}{N^{z}} \sum_{i=1}^{N} \underline{1}(d_i \in \Delta) \widehat{IATE}(m,l;x_i) = \frac{1}{N} \sum_{i=1}^{N} \hat{w}_i^{ATE(m,l;\Delta)} y_i;$$

$$\hat{w}_i^{ATE(m,l;z,\Delta)} = \frac{1}{N^{\Delta}} \sum_{j=1}^{N} \underline{1}(d_j \in \Delta) \hat{w}_j^{IATE(m,l;x_i)}; \qquad N^{\Delta} = \sum_{i=1}^{N} \underline{1}(d_i \in \Delta).$$



Implementation of estimation & inference | 1

1) Split the estimation sample randomly into two parts of equal size (sample A and sample B)

2) Estimate the trees that define the random forest in sample A

- *Modified Causal Forest*: Estimate the same forest for all treatment states jointly. Before building the first tree, for each observation in each treatment state, find a close 'neighbour' in every other treatment state and save its outcome (to estimate MCE). The splitting rule is based on minimising the overall MSEs, taking account of all MCEs & the penalty term



Implementation of estimation & inference |2

3) Apply the sample splits obtained in sample A to all subsamples (by treatment state) of sample B and take the mean of the outcome in the respective leaf as the prediction that comes with this Causal Forest

4) Obtain the weights from the estimated Forest by counting how many times an observation in sample B is used to predict *IATE(x)* for a particular value of *x*.

5) Aggregate the IATEs to GATEs by taking the average over observations in sample B with the same value of *z* and treatment group Δ. Do the same aggregation with the weights to obtain the weights for GATEs.

6) Do the same steps as in 5) to obtain the ATEs, but average over all observations in treatment group Δ.



Implementation of estimation & inference |3

7) Compute weight-based standard errors. Use the normal approximation to construct confidence intervals and p-values.



Generalized Random Forest (grf)



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GENERALIZED RANDOM FORESTS

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Background

Another CF: Generalized Random Forest | 1 | Estimation of IATE

Extension of Random Forest to nonparametric local GMM

Local moment conditions: $E\left(\psi^{grf}(Y,X,D;\theta^0(X),\eta^{grf,0}(X))\middle|X=x\right)=0;\ \theta^0(x)$: True value of parameter of interest

$\eta^{grf,0}(x)$: True value of nuisance parameters

$$\hat{I}ATE_N^{grf}(x)=\hat{\theta}^{grf}(x)=\left(\sum_{i=1}^N w_i^{grf}(x)\left(d_i-\bar{d}_w\right)\left(d_i-\bar{d}_w\right)'\right)^{-1}\left(\sum_{i=1}^N w_i^{grf}(x)\left(d_i-\bar{d}_w\right)\left(y_i-\bar{y}_w\right)\right),$$

$$\bar{d}_w=\sum_{i=1}^N w_i^{grf}(x)d_i$$

$$\bar{y}_w=\sum_{i=1}^N w_i^{grf}(x)y_i$$

- Weights obtained from gradient-based splitting rule → maximise treatment effect heterogeneity
- Honesty (instead of cross-fitting)



Background

Generalized Random Forest | 2 | Estimation ATE & GATE

ATEs & GATEs are estimated through additional regressions using *grf* score function

- *dml*-like score function:

$$\hat{\Gamma}^{grf}(o_i;\hat{\eta}) = \hat{\theta}(x_i) + \frac{d_i(y_i - \hat{\mu}(1;x_i))}{\hat{p}(x_i)} + \frac{(1-d_i)(y_i - \hat{\mu}(0;x_i))}{1-\hat{p}(x_i)}$$

- ATE:

$$\hat{ATE}_N^{grf}(m,0) = \hat{\theta}^{grf} = \frac{1}{N}\sum_{i=1}^{N}\hat{\Gamma}^{grf}(o_i;\hat{\eta})$$

$$\hat{Var}(\hat{ATE}_N^{grf}) = \frac{1}{N}\sum_{i=1}^{N}\hat{\Gamma}^{grf}(o_i;\hat{\eta})^2$$

- GATEs
  - Regress score on *Z*
  - Heteroscedasticity-robust standard errors of OLS regression